

THE SPECTRA ATTACHED TO COEFFICIENT SEQUENCES OF THE LEVEL 32 NEWFORM: TWO REMARKS AND AN EXPERIMENT

VERSION 3

CLAUDE

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“Claude” is a large language model developed by Anthropic. This document was prepared on 5 July 2026 at the request of Barry Brent, in connection with his draft [4], and is signed by the model at his request. It is *version 3*: Sections 1–4 are carried over from version 1 with only small edits; Section 5 is the version-2 rewrite, in which the preliminary hypothesis of version 1 (a 2-adic reading of a single-seed experiment) was retired after a twenty-fold replication (see §5.2); and Section 6, new in version 3, answers a question of Brent’s about regularities among the minimal moduli for the *full* expansions $h_n = a(n)$ across CM curves. All classical results are referenced; all computational claims were verified in Python during the sessions of 5 July 2026, subject to the caveats of §5.7. The mathematics should be checked independently before being relied upon.

ABSTRACT. In Brent’s draft *Oscillating spectra associated to certain sequences* [4], a sequence $\{h_n\}_{n \geq 0}$ with $h_0 = 1$ is converted, via a Newton-type identity, into a Hessenberg matrix J_n for each n , and the minimum modulus μ_n among the eigenvalues of J_n (or of a “deformed” matrix $J_n^{(c)}$) is plotted against n . We examine the two coefficient sequences that [4] attaches to the elliptic curve 32a1 of Cremona’s tables: $h_n = a(p_n + 1)$ and $h_n = a(p_n)$. The first sequence collapses the construction completely — the spectra are single points, so Figure 20C of [4] is a horizontal line for structural reasons traceable to complex multiplication. For the second, CM theory specifies the input exactly (zero set, values, distribution), making 32a1 a laboratory for the question of what drives the oscillation of μ_n . A replicated, controlled experiment (20 seeds per control, $n \leq 300$) retires the preliminary hypothesis of version 1 and supports a different, quantitative account: within the range tested, the dip-oscillation of μ_n is governed by the smallest-modulus zero $t_0 = re^{i\theta}$ of the generating function $H(t) = \sum h_n t^n$ — dips occur when t_0 is not real, with apparent period determined by θ through an aliasing formula and peak height approximately $\pi/(2r \sin \theta)$. The account classified 115 of 120 control runs correctly (the five failures all being near-ties between competing zeros), predicted periods to within 10% in 99% of oscillating runs, and passed an invariance-in- c test and a cross-curve test. It yields falsifiable predictions about figures of [4] not consulted here, recorded in §5.5. Version 3 adds Section 6 on the full expansions $h_n = a(n)$ across CM curves: there the generating function is literally $1 + f(\tau)$, so the governing zero corresponds to the highest solution of $f(\tau) = -1$ in the upper half-plane; the one theorem-level regularity (density-one vanishing of the undeformed minimal moduli, with Landau–Serre thinning of the complement) is recorded with references; and a six-curve comparison finds the governing zeros confined to a narrow band $0.66 \leq |t_0| \leq 0.78$ while exhibiting both oscillating and non-oscillating members, non-invariance under quadratic twist, and one resolved near-tie.

1. SETUP: BRENT’S CONSTRUCTION

We recall the machinery of [4] in the form we need; nothing in this section is ours. Let $\{h_n\}_{n \geq 0}$ be a sequence in a field of characteristic zero with $h_0 = 1$. Define $\{j_n\}_{n \geq 1}$ by the Newton-type recursion

$$(D1) \quad n h_n = \sum_{r=1}^n j_r h_{n-r} \quad (n \geq 1),$$

equivalently, in terms of the generating functions $H(t) = \sum_{n \geq 0} h_n t^n$ and $J(t) = \sum_{r \geq 1} j_r t^r$, by

$$(D2) \quad t H'(t) = H(t) J(t).$$

When h_n is the complete homogeneous symmetric function of degree n , (D1) is Newton’s identity and j_r is the r th power sum; see [14, Ch. I]. Following

[4], let $J_n(\bar{j})$ denote the $n \times n$ lower Hessenberg matrix

$$J_n(\bar{j}) = \begin{pmatrix} j_1 & -1 & 0 & \cdots & 0 \\ j_2 & j_1 & -2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ j_{n-1} & j_{n-2} & \cdots & j_1 & -(n-1) \\ j_n & j_{n-1} & \cdots & j_2 & j_1 \end{pmatrix},$$

so that ([4, Lemma 2.2]) $\det J_n(\bar{j}) = n! h_n$. The *deformation by c* of [4] prepends c to the sequence $\bar{j} = (j_1, j_2, \dots)$:

$$\overline{j^{(c)}} = (c, j_1, j_2, \dots), \quad J_n^{(c)} := J_n(\overline{j^{(c)}}),$$

and one writes $h^{(c)}$ for the sequence obtained from $\overline{j^{(c)}}$ by running (D1) forwards. Finally, μ_n denotes the minimum modulus among the eigenvalues of the matrix under discussion. The computational engine of [4] is the following consequence, via (D1), of a theorem of Vein and Dale [22, Thm. 4.23]:

Theorem 1.1 ([4, Thm. 2.3]). *With h and j related by (D1),*

$$\chi(J_n(\bar{j}))(x) = \sum_{r=0}^n \binom{n}{r} r! h_r (-1)^r x^{n-r}.$$

Two elementary observations about this setup will be used repeatedly.

Lemma 1.2. *Deformation by c acts on generating functions by $J(t) \mapsto ct + tJ(t)$. Consequently $H^{(c)}$ is the unique power-series solution with constant term 1 of*

$$\frac{d}{dt} \log H^{(c)}(t) = c + J(t),$$

where $J = tH'/H$ is attached to the undeformed sequence. In particular, if $h_n = 0$ for all $n \geq 1$ (so $J = 0$), then $H^{(c)}(t) = e^{ct}$ and $h_n^{(c)} = c^n/n!$.

Proof. Prepending c sends $j_r \mapsto j_{r-1}$ for $r \geq 2$ and puts $j_1^{(c)} = c$, which on generating functions is exactly $J \mapsto ct + tJ$. Substituting into (D2) and dividing by t gives the displayed equation. If $J = 0$ the equation is $(\log H^{(c)})' = c$. $\square$

Lemma 1.3. *For any sequence g with $g_0 = 1$ (in practice $g = h^{(c)}$), Theorem 1.1 may be rewritten*

$$\chi(J_n(\bar{j}_g))(x) = (-1)^n n! \sum_{m=0}^n g_{n-m} \frac{(-x)^m}{m!}.$$

In particular the characteristic polynomials are g -weighted partial sums of the exponential series in $-x$. Three special cases: if $g_n = c^n/n!$ the sum telescopes to $(x-c)^n$ by the binomial theorem; if $g = (1, 0, 0, \dots)$ only the $m = n$ term survives and $\chi(x) = x^n$; and if $g \equiv 1$ one obtains $(-1)^n n! s_n(-x)$, where s_n is the classical n th partial sum of the exponential series studied in

[21, 16]. Setting $x = 0$ recovers $\chi(0) = (-1)^n n! g_n = (-1)^n \det J_n$, consistent with [4, Lemma 2.2].

Proof. Substitute $m = n - r$ in Theorem 1.1 and use $\binom{n}{r} r! = n! / (n - r)!$. $\square$

Remark 1.4. Lemma 1.3 explains why the asymptotic theory of the zeros of the partial sums of e^z — Szegő [21], Newman–Rivlin [16] — is the natural comparison class for the spectra studied in [4]: the matrices J_n have been traded for perturbed exponential partial sums, with the sequence g read *backwards* through the factorially-decaying window $1/m!$. Since $1/m!$ decays so fast, the low-order coefficients of $\chi(J_n)(x)$ — which control the eigenvalues of small modulus, hence μ_n — are governed by a short sliding window $(g_n, g_{n-1}, g_{n-2}/2!, \dots)$ at the recent end of the sequence. The μ_n -plots of [4] are therefore plots of a *local* functional of the input sequence. Section 5 builds on this by asking what governs the asymptotics of that window.

2. THE LEVEL 32 NEWFORM AND ITS COEFFICIENTS

Cremona’s curve 32a1 is $E : y^2 = x^3 - x$, of conductor 32 [5, 13]. By modularity its L -series coincides with that of the unique newform $f_{32} \in S_2(\Gamma_0(32))$, which has the classical eta-quotient expression

$$(1) \quad f_{32}(\tau) = \eta(4\tau)^2 \eta(8\tau)^2 = q \prod_{n=1}^{\infty} (1 - q^{4n})^2 (1 - q^{8n})^2 = q - 2q^5 - 3q^9 + 6q^{13} + \dots,$$

where $q = e^{2\pi i \tau}$; see Martin–Ono [15, Thm. 2], whose table lists all elliptic curves whose newforms are eta quotients (conductors 27, 32, 36, 48, 64, 80, 108, 144 appear), and [13], newform 32.2.a.a. We write $a(m)$ for the coefficient of q^m , normalized with $a(1) = 1$, so that $a(p) = p + 1 - \#E(\mathbb{F}_p)$ for primes p of good reduction [7, Ch. 8].

The curve E has complex multiplication by the Gaussian integers $\mathbb{Z}[i]$, and f_{32} is the theta series of a Hecke Grössencharacter of $\mathbb{Q}(i)$ of conductor $(1 + i)^3$ [10, 11]. Everything we need about the coefficients is collected in the following proposition. Parts (i)–(iv) are classical; part (v) is elementary linear algebra over $\mathbb{F}_p$ dressed as group theory.

Proposition 2.1. *Let $a(m)$ be as above. Then:*

- (i) (Support.) $a(m) = 0$ unless $m \equiv 1 \pmod{4}$.
- (ii) (Supersingular vanishing.) $a(2) = 0$, and $a(p) = 0$ for every prime $p \equiv 3 \pmod{4}$; these are exactly the supersingular primes of E , by Deuring’s criterion [6] (a prime inert in the CM field is supersingular).
- (iii) (Gauss’s formula.) For $p \equiv 1 \pmod{4}$, write $p = a^2 + b^2$ with a odd and the signs normalized so that $a + b \equiv 1 \pmod{4}$; then $a(p) = 2a$. (See [11, Ch. 18] for the classical statement, which goes back to

Gauss's work on biquadratic residues; the normalization stated here was verified against point counts for all $p < 3000$.)

- (iv) (Value distribution.) Write $a(p) = 2\sqrt{p}\cos\theta_p$ for good p . Then $\theta_p = \pi/2$ exactly on the density- $\frac{1}{2}$ set of inert primes, while on the split primes θ_p equidistributes with respect to the uniform measure on $[0, \pi]$; equivalently, $a(p)/2\sqrt{p}$ follows the arcsine density $\frac{1}{\pi\sqrt{1-x^2}}$ on the split primes. This is Hecke's equidistribution theorem for Grössencharacter angles [9, 10]; it is the CM case of the Sato–Tate law and differs from the semicircle distribution governing non-CM curves [3].
- (v) (2-adic stratification.) E has full rational 2-torsion: $E(\mathbb{Q})[2] = \{O, (0, 0), (\pm 1, 0)\} \cong (\mathbb{Z}/2)^2$. Hence for every odd prime p (all of which have good reduction), reduction injects $E(\mathbb{Q})[2]$ into $E(\mathbb{F}_p)$ [20, VII.3.1], so $4 \mid \#E(\mathbb{F}_p)$ and

$$a(p) \equiv p + 1 \pmod{4}.$$

Combining with (ii) and (iii): every nonzero coefficient $a(p)$ satisfies $a(p) \equiv 2 \pmod{4}$, i.e., has 2-adic valuation exactly 1 (consistent with $a(p) = 2a$, a odd), and the coefficients at $p \equiv 3 \pmod{4}$, which (v) alone would only force to be $\equiv 0 \pmod{4}$, are exactly 0 by (ii).

Proof of (i). Every monomial produced by the product in (1) has exponent divisible by 4; multiplying by the leading q makes every exponent $\equiv 1 \pmod{4}$. (Alternatively: (ii), multiplicativity, and the Hecke recursion $a(p^k) = a(p)a(p^{k-1}) - pa(p^{k-2})$ [7, Ch. 5] show that the prime powers in the support are $\equiv 1 \pmod{4}$, and the support is closed under coprime multiplication.) Forms with such thin support are *lacunary*; by a theorem of Serre [19] an integral-weight holomorphic newform is lacunary if and only if it has CM. $\square$

3. TOTAL COLLAPSE: THE SEQUENCE $h_n = a(p_n + 1)$

Proposition 3.1. $a(p + 1) = 0$ for every prime p .

Proof. If p is odd then $p + 1$ is even, hence $\not\equiv 1 \pmod{4}$; and $2 + 1 = 3 \equiv 3 \pmod{4}$. Apply Proposition 2.1(i). $\square$

Corollary 3.2. Let $h_0 = 1$ and $h_n = a(p_n + 1)$, so $h_n = 0$ for all $n \geq 1$. Then:

- (i) $j_n = 0$ for all n ; the undeformed $J_n(\bar{j})$ is strictly upper triangular, hence nilpotent, with $\chi(x) = x^n$, spectrum $\{0\}$, and $\mu_n \equiv 0$.
- (ii) For any c , $J_n^{(c)} = cI_n + N_n$ with N_n the strictly upper triangular matrix carrying $-1, -2, \dots, -(n-1)$ on the superdiagonal; hence

$$\chi(J_n^{(c)})(x) = (x - c)^n,$$

the spectrum is the single point $\{c\}$, and $\mu_n \equiv |c|$.

Proof. (i) By induction in (D1): $n \cdot 0 = j_n h_0$. Every entry of $J_n(\bar{j})$ on or below the diagonal is some $j_k = 0$. (ii) $\bar{j}^{(c)} = (c, 0, 0, \dots)$, so the diagonal is constant c and the subdiagonals vanish; a triangular matrix has characteristic polynomial determined by its diagonal. Alternatively, by Lemma 1.2, $h_r^{(c)} = c^r / r!$, and Theorem 1.1 gives $\chi(x) = \sum_r \binom{n}{r} (-c)^r x^{n-r} = (x-c)^n$. $\square$

Remark 3.3. (a) Figure 20C of [4] plots μ_n for this sequence with $c = 1$: by Corollary 3.2(ii) it is the constant function $\mu_n = 1$, and the stored characteristic polynomials $(x-1)^n$ — whose coefficient arrays are the rows of Pascal’s triangle with alternating signs — are binomial expansions and nothing more. The only fact about f_{32} that survives into Figure 20C is Proposition 3.1. Any sequence with $h_n = 0$ for $n \geq 1$ produces identical output.

(b) The same collapse occurs verbatim for Cremona curve 36a1, whose newform is $\eta(6\tau)^4$ [15], supported on exponents $\equiv 1 \pmod{6}$; since $p+1 \equiv 1 \pmod{6}$ would force $6 \mid p$, again $a(p+1) \equiv 0$.

(c) Curve 27a1 is a near miss. Its newform $\eta(3\tau)^2 \eta(9\tau)^2$ [15] is supported on exponents $\equiv 1 \pmod{3}$, and $p+1 \equiv 1 \pmod{3}$ forces $p = 3$; the only surviving term is $a(4) = -2$. Thus $h = (1, 0, -2, 0, 0, \dots)$, $H(t) = 1 - 2t^2$ exactly, (D2) gives $j_{2k} = -2^{k+1}$ and $j_{2k+1} = 0$, and Theorem 1.1 yields the closed form

$$\chi(J_n(\bar{j}))(x) = x^n - 2n(n-1)x^{n-2} = x^{n-2}(x^2 - 2n(n-1))$$

for the undeformed matrices: spectrum $\{0\} \cup \{\pm\sqrt{2n(n-1)}\}$, so $\mu_n = 0$ for $n \geq 3$, and the deformed spectra are governed by the single coefficient $a(4)$.

(d) Consequently the sequences $h_n = a(p_n + 1)$, introduced in [4] as controls for the sequences $h_n = a(p_n)$, are uninformative for the CM curves 32a1 and 36a1 (and nearly so for 27a1); their plots should be excluded from any tally of oscillating versus non-oscillating data sets, and in particular from tests of closure properties of the proposed spaces V_h , since a constant sequence is “coherent” only vacuously. For curves without CM — e.g. 11a1 or 33a1 — the coefficients $a(p_n + 1)$ form a genuinely nonvanishing, sign-varying sequence and the control carries real content.

4. THE OPPOSITE EXTREME: THE SEQUENCE $h_n = a(p_n)$

For $h_n = a(p_n)$, CM theory does not collapse the construction; instead it specifies the input sequence completely. We record the consequences for the machinery of Section 1.

Corollary 4.1. *Let $h_0 = 1$ and $h_n = a(p_n)$ for the curve 32a1. Then:*

- (i) (Zero set known exactly.) $h_n = 0$ if and only if $p_n = 2$ or $p_n \equiv 3 \pmod{4}$. The Lehmer-type nonvanishing question that motivates the study of $\tau(p_n)$ in [4] is therefore trivial for this sequence, and its zero set has density $\frac{1}{2}$ by Dirichlet.
- (ii) (Forced singularities.) Since $\det J_n(\bar{j}) = n! h_n$, the undeformed matrix J_n is singular ($\mu_n = 0$) precisely at the density- $\frac{1}{2}$ set of n in

- (i); every undeformed μ_n -plot for this sequence returns to 0 on a set of indices governed by the distribution of primes in the progression $3 \bmod 4$.
- (iii) (Exact computability.) By Proposition 2.1(iii), h_n is computable exactly in $O(\sqrt{p_n})$ integer operations (or $O(\log p_n)$ via Cornacchia's algorithm applied to $p_n = a^2 + b^2$), with no point counting, no modular symbols, and no floating point — of practical relevance to the precision issues documented in [4].
- (iv) (Statistics known.) The nonzero values, normalized by $2\sqrt{p_n}$, follow the arcsine law (Proposition 2.1(iv)); numerically, for $p < 20000$ the vanishing fraction is 0.5027 and the Kolmogorov–Smirnov distance of the normalized nonzero values to the arcsine law is 0.014, against 0.173 to the semicircle law.
- (v) (2-adic rigidity.) Every term of the sequence is either 0 or $\equiv 2 \pmod{4}$ (Proposition 2.1(v)).

Items (i)–(iv) say that the “arithmetic characteristics of the inputs,” in the language of [4, Remark on controls], are theorems rather than data for this curve. This makes 32a1 the ideal laboratory for the question raised there: *which* characteristics of the input drive the observed oscillation of μ_n ? One can manufacture synthetic sequences matching any chosen subset of (i), (iv), (v) and compare spectra. That experiment is the subject of the next section.

5. A REPLICATED EXPERIMENT ON FIGURE 20B

Version 1 of these notes reported a single-seed experiment at $n \leq 140$ and read off a preliminary hypothesis: that the dip-oscillation of μ_n tracked the 2-adic valuations of the nonzero h_n . The present section expands that experiment twenty-fold and to $n \leq 300$ (the range of the figures in [4]). The preliminary hypothesis did not survive; the expanded data instead support a different account, stated as Heuristic 5.1 and tested in §5.4.

5.1. Method and validation. All pipelines are exact until the final root extraction. From integer h , (D1) produces integer j ; c is prepended; and the integers $D_m := m! h_m^{(c)}$ are computed by the integer recursion $D_m = \sum_{r=1}^m j_r^{(c)} D_{m-r} (m-1)! / (m-r)!$ obtained from (D1). The characteristic polynomial is assembled in the normalization of Lemma 1.3, whose coefficient of x^m is $(-1)^m D_{n-m} / ((n-m)! m!)$.

Two numerical devices are then applied, each validated. (a) *Truncation.* Since the coefficients decay through the factor $1/m!$, the polynomial is truncated at degree $M = 100$ (fallback $M = 140$), and only roots of modulus $\leq R = 22$ (fallback 38) are trusted; the discarded tail contributes at most $\max_k |h_k^{(c)}| \cdot R^M / M!$ on the disk $|x| \leq R$, which is $< 10^{-16}$ times the polynomial's scale in every regime encountered. If no root lies inside the fallback

radius, μ_n is recorded as exceeding it (“runaway” at that index). (b) *Scale-invariant normalization*. The exact rational coefficients can overflow double precision for synthetic sequences; they are therefore converted in mantissa–exponent form and rescaled by a common power of 2, which leaves the roots unchanged.

Validation: (1) on the real sequence the pipeline agrees with the independent exact-rational implementation of version 1 to 2×10^{-13} uniformly for $n \leq 140$; (2) at $n = 150, 220, 300$ the double-precision values of μ_n agree with 60-digit multiprecision recomputations to six decimal places (0.373487, 1.779994, 2.230402 respectively); (3) the real sequence reproduces the qualitative features of Figure 20B of [4]: oscillation onset near $n \approx 50$, maxima ≈ 3.6 – 3.7 , minima near 0.

5.2. Replication of the version-1 experiment. Synthetic controls match selected statistics of the real sequence $h_n = a(p_n)$: type A keeps the real zero pattern (zeros at $p_n = 2$ and $p_n \equiv 3 \pmod{4}$) and replaces the nonzero values by independent arcsine-distributed integers of the correct scale; A2, A0, A1 are the same with the values forced $\equiv 2 \pmod{4}$, $\equiv 0 \pmod{4}$, and odd, respectively; type B keeps the real nonzero values in order and randomizes the zero *positions*; type C randomizes everything. Twenty seeds of each were run at $N = 300$, $c = 1$. Writing $\text{dip} = \min_{100 \leq n \leq 300} \mu_n$, the medians and interquartile ranges over seeds:

type	dip, median [IQR]	mean of μ_n	runaway seeds (of 20)
real 32a1	0.02	1.81	—
B	0.01 [0.01, 0.04]	2.25	3
A2	0.02 [0.01, 0.22]	1.54	5
A1	0.01 [0.00, 0.04]	1.52	4
A	0.02 [0.01, 0.04]	1.50	4
A0	0.04 [0.01, 0.23]	1.63	5
C	16.4 [0.01, 36.9]	27.3	12

(“Runaway” seeds are those with $\text{dip} \geq 0.3$; in such runs μ_n drifts upward without approaching 0.) Two conclusions. First, the version-1 reading fails to replicate: at $n \leq 300$, most seeds of *every* type — including A and A0, whose suppression at $n \leq 140$ motivated the 2-adic hypothesis — dip and oscillate; the apparent discrimination by residue class was a transient and/or small-sample effect, and the hypothesis is retired. Second, the data are bimodal: each type contains a minority of runaway seeds, type C most often. Any correct account must predict, seed by seed, which runs dip and which run away.

5.3. A heuristic mechanism. By Remark 1.4, μ_n is controlled by the recent window of $h^{(c)}$, so by the asymptotics of $h_k^{(c)}$; and by Lemma 1.2 the singularities of $H^{(c)}$ are located at the *zeros of H* : near a simple zero t_0 of H one has $tH'/H \sim t_0/(t - t_0)$, whence $H^{(c)}(t) = (1 - t/t_0)^{t_0}$.

(analytic, nonvanishing), a branch point at t_0 of exponent t_0 . (The deformation parameter c contributes only an entire factor and moves nothing.) Suppose the smallest-modulus zeros of H inside its disk of convergence form a simple conjugate pair $t_0, \bar{t}_0 = re^{\pm i\theta}$, with the next-smallest zero of strictly larger modulus. Singularity analysis [8] then gives $h_k^{(c)} \sim 2 \operatorname{Re}[C_k t_0^{-k}]$ with C_k of slowly varying modulus and phase, and the window sum of Lemma 1.3 telescopes against the exponential series:

$$\sum_{m \geq 0} h_{n-m}^{(c)} \frac{(-x)^m}{m!} \approx 2 \operatorname{Re} [C_n t_0^{-n} e^{-t_0 x}] = 2|C_n| r^{-n} e^{-(r \cos \theta)x} \cos(\varphi_n - (r \sin \theta)x),$$

with $\varphi_n = \arg C_n - n\theta$. For real x this vanishes on an arithmetic progression of spacing $\pi/(r \sin \theta)$ that translates by $\theta/(r \sin \theta)$ as $n \rightarrow n+1$. This suggests, within its range of validity:

Heuristic 5.1. Let $t_0 = re^{i\theta}$ be the smallest-modulus zero of $H(t) = \sum h_n t^n$ (computed, in practice, from a high-degree truncation), assumed simple and strictly smaller in modulus than all other zeros. Then, for n large:

- (i) (Dip criterion.) μ_n dips toward 0 and oscillates if t_0 is not real ($0 < \theta < \pi$); if t_0 is real (positive or negative), the window has no real zeros near the origin and μ_n runs away.
- (ii) (Aliased period.) The zeros sweep past the origin with period π/θ in n ; sampled at integer n , the apparent period is $1/\min(f, 1-f)$ with $f = \theta/\pi$.
- (iii) (Peak height.) The peaks of μ_n approach half the zero spacing, $\pi/(2r \sin \theta)$.
- (iv) (Invariance in c .) None of (i)–(iii) depends on c .
- (v) (Onset.) The behavior sets in once the t_0 -term dominates the competing singularities, roughly when $(r_2/r_1)^n$ is large, where r_2/r_1 is the ratio of the two smallest zero moduli.

We emphasize that Heuristic 5.1 is a heuristic: the derivation ignores the slowly-varying factor C_k (whose phase drifts like $-\operatorname{Im}(t_0) \log k$, predicting slow period drift), assumes a single dominant simple zero, and says nothing when H has no zeros inside its disk of convergence. It is stated in order to be tested.

5.4. Tests. (1) *The real sequence.* The smallest zero of the degree-300 truncation of $\sum a(p_n)t^n$ is $t_0 = 0.7053 e^{\pm 2.4748i}$. Then $f = \theta/\pi = 0.7877$, so the predicted apparent period is $1/(1-0.7877) = 4.71$ against the measured 4.8; the predicted peak is $\pi/(2 \cdot 0.7053 \cdot \sin 2.4748) = 3.60$ against the measured maximum 3.61 over $100 \leq n \leq 300$. The zero-modulus ratio $r_2/r_1 = 1.039$ makes the t_0 -term dominate on a scale of some tens of steps, consistent with the observed onset near $n \approx 50$.

(2) *Seed-by-seed classification.* Across the 120 control runs of §5.2, classifying a run as “dip” when $\text{dip} < 0.3$ and its zero as “complex” when $0.02 < \theta < \pi - 0.02$:

	smallest zero complex	smallest zero real
run dips	84	3
run runs away	2	31

i.e., 115/120 runs classified correctly. All five misclassified runs have zero-modulus ratio $r_2/r_1 \leq 1.05$ — near-ties between competing zeros, which is precisely where the single-dominant-zero assumption of Heuristic 5.1 fails; no misclassification occurs with $r_2/r_1 > 1.05$.

(3) *Periods.* Among the 84 correctly-predicted dipping runs, the aliased period of Heuristic 5.1(ii) matches the measured dominant period with median relative error 0.8%; 99% of runs agree within 10%.

(4) *Invariance in c .* For the real sequence with $c \in \{1, 2, 3, 5\}$: measured dips 0.016, 0.038, 0.001, 0.010; means 1.81, 1.80, 1.80, 1.75; measured period 4.8 in all four cases, as predicted. (Note that this test would have refuted a mechanism in which the residue classes of the values interact with $c = 1$ specifically.)

(5) *Cross-curve test.* For three further sequences $h_n = p_n + 1 - \#C(\mathbb{F}_{p_n})$ (naive point counts, $n \leq 300$, $c = 1$; at the finitely many bad primes the same counting formula is used, so the sequences are well-defined even where they differ from newform coefficients):

curve	smallest zero	r_2/r_1	prediction	observed
32a1: $y^2 = x^3 - x$	$0.705 e^{\pm 2.475i}$	1.04	dips, period 4.7, peak 3.6	dips, 4.8, 3.61
$y^2 = x(x-1)(x+2)$	-0.614	1.01	runaway	no dips (min 2.1)
11a1	+0.430	1.82	runaway	runaway ($\mu_n > 38$)
37a1: $y^2 + y = x^3 - x$	+0.311	2.51	runaway	runaway ($\mu_n > 38$)

The second curve is non-CM (its j -invariant 21952/9 is not an algebraic integer) with full rational 2-torsion; under the retired 2-adic hypothesis its mixed strata and the mixed strata of 11a1 and 37a1 would not have been distinguishable from one another, whereas here each outcome is predicted by its own zero. We note also that the location of the smallest zero on the *negative* real axis for the second curve, versus positive for 11a1 and 37a1, is invisible in μ_n (both give runaway) but visible in the sign pattern of $h_k^{(c)}$.

5.5. Falsifiable predictions. The following concern figures of [4] that were *not consulted* in preparing these notes (only their captions were); they can be checked against the stored data of [4].

(P1) Figure 8B (11a1, $h_n = a(p_n)$, $c = 1$) should show μ_n growing without near-zero dips.

(P2) Figure 2 ($h_n = \lambda_n = \tau(p_n)$, $c = 1$): the smallest zero of the degree-300 truncation of $\sum \tau(p_n)t^n$ is $0.0435 e^{\pm 0.764i}$ ($r_2/r_1 = 1.79$),

predicting dips, apparent period ≈ 4.1 , and peak height ≈ 52 . Our reimplementaion measures period 3.7 and maximum 51.4 over $100 \leq n \leq 300$; the plotted Figure 2 should agree within the drift allowed by the C_k factor.

- (P3) Figure 20A (32a1, $h_n = a(n)$, $c = 1$): smallest zero $0.7125 e^{\pm 2.7245i}$ ($r_2/r_1 = 1.17$), predicting dips, apparent period ≈ 7.5 , peak ≈ 5.4 ; our reimplementaion measures 7.7 and 5.45.
- (P4) For any sequence in the catalog of [4], redrawing the plot with a different c should leave the presence of dips, the period, and the peak height essentially unchanged.

5.6. What this does and does not explain. Within the range tested, Heuristic 5.1 converts the question “which sequences h oscillate?” into “where is the smallest zero of $H(t) = \sum h_n t^n$?”. This is consistent with the report in [4] that no invariant of the underlying modular forms was found to predict oscillation: the criterion is analytic, and how the arithmetic of a modular form governs the location of the smallest zero of $\sum a(p_n) t^n$ is, to our knowledge, open — the mystery is relocated, not resolved. Two further remarks. First, the proposal of [4] that coherent sequences might form vector spaces V_h acquires a concrete difficulty from this viewpoint: coherence is governed by the smallest zero of H , and the zeros of $H_1 + H_2 - 1$ are not in any evident way controlled by those of H_1 and H_2 , so additive closure would itself be a phenomenon in need of explanation wherever it is observed. Second, for the Lehmer sequence $\lambda_n = \tau(p_n)$, the mechanism ties the oscillation of [4]’s Figures 2–4 to the zero $0.0435 e^{\pm 0.764i}$ of the truncated $\sum \tau(p_n) t^n$; whether that zero (and hence the oscillation) persists as the truncation degree grows, and what if anything it means arithmetically, we do not know.

5.7. Caveats.

Caveat 5.2. (a) Root extraction is double-precision, validated against multiprecision only at three indices of the real sequence; the runaway values reported as “ $\mu_n > 38$ ” are truncation-radius statements, not computed minima. (b) The “zeros of H ” are zeros of a degree-300 truncation. For the sequences here, $|h_n|$ grows polynomially, so H has radius of convergence 1 and, by Hurwitz’s theorem [1, Ch. 5], zeros of truncations converge to zeros of H strictly inside the unit disk; all zeros used satisfy $r < 0.72$. The stability of these zeros under increase of the truncation degree was not systematically studied here. (c) Heuristic 5.1 assumes a simple, strictly dominant zero; the five misclassifications in §5.4(2) show its failure mode. (d) Period measurement uses a coarse discrete Fourier estimate on $n \in [100, 300]$. (e) Twenty seeds per control is adequate for the contingency table but modest for tail behavior. (f) These are experiments and heuristics; no statement in Section 5 beyond the validations of §5.1 is proved.

6. THE FULL EXPANSIONS $h_n = a(n)$ ACROSS CM CURVES

Brent asked whether known regularities exist among the minimal moduli for the plots with $h_n = a(n)$ — the full expansions, not merely the prime-indexed subsequences — across CM curves. To our knowledge no literature addresses these spectra (the construction is that of [4]), so “known” can only mean: consequences of the classical structure theory of CM forms, plus what Heuristic 5.1 makes of that structure. This section records one regularity that is a theorem (§6.2), one exact reinterpretation special to the full expansions (§6.1), and a six-curve computed comparison (§6.3) whose observed regularities and non-regularities are summarized in §6.4.

6.1. A modular reinterpretation of the governing zero. The full expansions occupy a special position in the framework of Section 5: their generating function *is* the modular form.

Lemma 6.1. *Let $f = \sum_{n \geq 1} a(n)q^n$ be any newform and let $h_0 = 1$, $h_n = a(n)$. Then, as analytic functions on the open unit disk under the substitution $t = q = e^{2\pi i \tau}$,*

$$H(t) = 1 + f(\tau), \quad J(t) = \frac{q \frac{d}{dq} f}{1 + f} = \frac{1}{2\pi i} \frac{d}{d\tau} \log(1 + f(\tau)).$$

Consequently the zeros of H in the unit disk correspond bijectively to the solutions of

$$f(\tau) = -1, \quad \tau \in \mathbb{H} \pmod{\mathbb{Z}},$$

via $t_0 = e^{2\pi i \tau_0}$, with $|t_0| = e^{-2\pi \operatorname{Im} \tau_0}$; the smallest-modulus zero of H — the input to Heuristic 5.1 — corresponds to the highest solution of $f(\tau) = -1$ in the upper half-plane. Moreover, since the $a(n)$ are real, zeros of H off the real axis come in conjugate pairs, and a zero on the real t -axis corresponds to a solution with $\operatorname{Re} \tau_0 \in \{0, \frac{1}{2}\}$, the two vertical lines on which q is real and f is real-valued. The dip criterion of Heuristic 5.1(i) therefore reads: the deformed μ_n -plot for $h_n = a(n)$ oscillates if and only if the highest solution of $f(\tau) = -1$ lies off the lines $\operatorname{Re} \tau \in \frac{1}{2}\mathbb{Z}$ (assuming the solution is simple and strictly dominant).

Proof. The identities are the definitions read under $t = q$; the correspondence of zeros is immediate, and $1 + f$ is real on $\operatorname{Re} \tau \in \frac{1}{2}\mathbb{Z}$ because the coefficients are real and q is real there. Note $f(\tau) \rightarrow 0$ as $\operatorname{Im} \tau \rightarrow \infty$, so solutions of $f = -1$ have bounded height and a highest solution exists locally; we do not claim uniqueness in general. $\square$

We are not aware of literature on the level sets $f(\tau) = c$ for $c \neq 0$ (the value distribution of newforms in the disk), in contrast to the well-developed theory of their zeros; Lemma 6.1 says the spectra of [4] for full expansions are, heuristically, probes of exactly this.

6.2. The theorem-level regularity: undeformed plots. For CM forms — precisely the newforms that are theta series of Hecke Grössencharacters [10, 17] — the full expansion is lacunary, and this forces a regularity of the *undeformed* minimal moduli that holds for every CM curve and is not experimental:

Proposition 6.2. *Let f be a CM newform with coefficients $a(n)$, and let μ_n be the minimal eigenvalue modulus of the undeformed J_n built from $h_n = a(n)$. Then $\mu_n = 0$ on a set of density one; more precisely,*

$$\#\{n \leq x : \mu_n \neq 0\} = O\left(\frac{x}{\sqrt{\log x}}\right).$$

Proof. $\mu_n = 0$ iff $\det J_n = n!a(n) = 0$. For CM forms, $\#\{n \leq x : a(n) \neq 0\} \sim c_f x / \sqrt{\log x}$ for a positive constant c_f ; this is due to Serre [18, §7] (the qualitative density-zero statement characterizes CM among newforms [19]). For f_{32} the count is of the same nature as Landau’s classical asymptotic for the integers representable as a sum of two squares [12], since $a(n) \neq 0$ requires every inert prime to divide n to an even power. $\square$

So for every CM curve the undeformed full-expansion plot is pinned to 0 on almost all indices, and the exceptional set thins like $1/\sqrt{\log x}$ — a regularity, though a slow-motion one: at $x = 600$ the observed nonzero fractions for the six curves below still range from 0.13 to 0.27. The interesting comparisons are therefore in the deformed plots, where Proposition 6.2 is no obstruction and Heuristic 5.1 applies.

6.3. Six CM curves compared. We assembled $a(n)$ for $n \leq 600$ for six CM curves: the three eta-quotient curves 27a1 ($\eta(3\tau)^2\eta(9\tau)^2$, CM by $\mathbb{Q}(\sqrt{-3})$), 32a1, and 36a1 ($\eta(6\tau)^4$, CM by $\mathbb{Q}(\sqrt{-3})$) from their product expansions [15]; and, from point counts extended multiplicatively by the Hecke recursion, the models $y^2 + xy = x^3 - x^2 - 2x - 1$ (49a1, CM by $\mathbb{Q}(\sqrt{-7})$), $y^2 = x^3 - 4x$ (the quadratic twist of 32a1 by 2, conductor 64), and $y^2 + y = x^3 - x^2 - 7x + 10$ (CM by $\mathbb{Q}(\sqrt{-11})$); the curve of conductor 121 usually labeled 121b1, though we verified the CM property, not the conductor). Validations: the eta expansions agree with point counts at all good primes $p < 500$; for all six curves the vanishing $a(p) = 0$ occurs exactly at the primes inert in the CM field for all good $p < 1000$ (Deuring [6]); the support congruences hold for all $n \leq 600$ (nonzero $a(n)$ have $n \equiv 1 \pmod{3}$, 4, 6 for conductors 27, 32, 36; $n \in \{1, 2, 4\} \pmod{7}$ for 49; $n \in \{1, 3, 4, 5, 9\} \pmod{11}$ for 121); and the twist identity $a_{64}(n) = \left(\frac{2}{n}\right) a_{32}(n)$ holds exactly for all $n \leq 600$. (The $j = 0$ pair 27a1, 36a1 is related by *sextic* twisting, so no ± 1 relation between their coefficients is expected; both families are treated classically via Jacobi sums in [11, Ch. 18].)

For each curve, the smallest zero of the degree-300 truncation of $H(t) = 1 + \sum a(n)t^n$ was computed and found stable to four decimal places under doubling the truncation degree to 600; μ_n was then computed for $n \leq 300$,

$c = 1$, by the pipeline of §5.1. Writing $t_0 = re^{i\theta}$ and $\tau_0 = \frac{\theta}{2\pi} + i \frac{-\log r}{2\pi}$ for the corresponding solution of $f = -1$:

curve	$t_0 = re^{i\theta}$	r_2/r_1	predicted	meas. period	meas. peak	dip
27a1	$0.6602 e^{i\pi}$ (real, < 0)	1.19	runaway	—	28.5, rising	20.0
32a1	$0.7125 e^{\pm 2.7245i}$	1.17	7.5, peak 5.4	7.7	5.45	0.003
36a1	$0.7175 e^{\pm 2.7833i}$	1.14	8.8, peak 6.2	8.7	6.18	0.070
49a1	$0.7723 e^{\pm 1.6306i}$	1.02	2.1, peak 2.0	2.1	2.50	0.006
64	$0.7540 e^{\pm 3.0349i}$	1.04	29.4, peak 19.6	28.6	16.2	0.099
121	0.7485 (real, > 0)	1.01	runaway*	7.7	3.19	0.012

The conductor-121 row, marked *, is a live instance of the near-tie failure mode identified in §5.4(2): its smallest zero is exactly real (so Heuristic 5.1 taken literally predicts runaway), but the ratio to the next zero is only 1.013, and that next zero is the complex pair $0.7583 e^{\pm 2.7296i}$, whose aliased period is 7.62 — against the measured 7.7. The subdominant pair drives the observed oscillation, with the peak (3.19, against the pair’s solo prediction 5.17) plausibly damped by the competing zero-free contribution of the real zero. By contrast 27a1’s real zero is well separated ($r_2/r_1 = 1.19$) and the plot runs away as predicted. In the τ -coordinate of Lemma 6.1, the six highest solutions of $f(\tau) = -1$ are

$$\begin{aligned} 27: 0.5000 + 0.0661i, \quad 32: 0.4336 + 0.0540i, \quad 36: 0.4430 + 0.0528i, \\ 49: 0.2595 + 0.0411i, \quad 64: 0.4830 + 0.0449i, \quad 121: 0.0000 + 0.0461i, \end{aligned}$$

with the two real-axis zeros sitting exactly on the lines $\operatorname{Re} \tau \in \{0, \frac{1}{2}\}$, as Lemma 6.1 requires.

6.4. Observed regularities and non-regularities. On the evidence of six curves — a small sample, and everything below is empirical except where referenced:

- (R1) *A common scale.* All six governing zeros have modulus in the narrow band $0.66 \leq r \leq 0.78$ (equivalently, all six highest solutions of $f = -1$ have $0.041 \leq \operatorname{Im} \tau_0 \leq 0.067$). Consequently the full-expansion plots operate at comparable, modest scale — peaks of order 2–20, periods of order 2–30 — in sharp contrast to the prime-indexed Lehmer sequence, whose governing zero $0.0435 e^{\pm 0.764i}$ (§5.5) produces peaks near 52. We see no reason the band should be sharp, but its narrowness across four CM fields (discriminants $-3, -4, -7, -11$) and both twist-related pairs is the most uniform feature observed.
- (R2) *Not all CM full expansions oscillate.* 27a1 runs away; its highest solution of $f = -1$ lies on $\operatorname{Re} \tau = \frac{1}{2}$. Whether a given CM curve’s plot oscillates appears to be decided not by CM structure per se but by whether that highest solution falls on the real- q lines — which happened here in two of six cases, in two different ways ($\operatorname{Re} \tau_0 = \frac{1}{2}$ dominant for 27; $\operatorname{Re} \tau_0 = 0$ in a near-tie for 121).

- (R3) *Minimal moduli are not twist-invariant.* The pair $32a1 \rightarrow 64$ differs only by the character $\left(\frac{2}{\cdot}\right)$ on coefficients, yet the governing zero moves from $0.7125 e^{\pm 2.7245i}$ to $0.7540 e^{\pm 3.0349i}$ and the period from ≈ 7.5 to ≈ 29 . Since quadratic twisting preserves essentially every coarse invariant of the form (field of coefficients, CM field, $|a(p)|$ for odd p), this is a sharp illustration — within the heuristic — of why no such invariant was found in [4] to predict the plots: the governing quantity is a value-locus datum of f , not an invariant of its Galois representation up to twist.
- (R4) *No support-modulus pattern.* $32a1$ and 64 share the support congruence $n \equiv 1 \pmod{4}$ but have quite different θ ; the support congruences alone do not locate the zero.

Natural extensions we have not carried out: the remaining CM conductors over $\mathbb{Q}$ (e.g. 256, with CM by $\mathbb{Q}(\sqrt{-2})$, and the curves attached to discriminants $-19, -43, -67, -163$ at conductors 361, 1849, 4489, 26569), which would test (R1) beyond the smallest conductors (all thirteen rational CM j -invariants have class number one, so class number does not vary in this family); truncation degrees beyond 600; and the deformed plots for the *prime-indexed* sequences of the same six curves, for which Corollary 4.1-style exact input theorems are available in each case. For a new curve the protocol is: take a minimal Weierstrass model from [13]; compute $a(p)$ by point counting for p up to the 600th prime, 4409, setting $a(p) = 0$ at the single ramified prime; extend multiplicatively by the Hecke recursion (Appendix A); validate that $a(p) = 0$ exactly at the primes inert in the CM field for all good $p < 1000$ [6], and that the nonzero $a(n)$, $n \leq 600$, lie in the norm classes of the field; locate the smallest zero of the truncated H at degrees 300 and 600 and check its stability; and only then compare the predictions of Heuristic 5.1 with the μ -pipeline at $n \leq 300$, $c = 1$. Since the band of (R1) was observed rather than predicted, the honest test is whether the new zero moduli continue the clustering, not a sharp pass/fail against the interval $[0.66, 0.78]$.

ACKNOWLEDGMENT AND SIGNATURE

These notes respond to questions posed by Barry Brent on 5 July 2026 about his draft [4]; the construction studied here is his, as are the data sets that prompted the analysis. Version 2 exists because he asked that the experiment of version 1 be expanded before its conclusions were trusted; the expansion overturned those conclusions, which is the reason one expands experiments. Version 3 responds to his further question whether known regularities exist among the minimal moduli for the full expansions $h_n = a(n)$ of CM curves. Errors introduced in the writing are mine.

— CLAUDE (Anthropic)
6 July 2026, version 3

APPENDIX A. REFERENCE IMPLEMENTATION

The following Python excerpt implements the pipeline of §5.1: exact coefficients $a(p)$ for 32a1, the exact integer (D1) pipeline with deformation, truncated root extraction with scale-invariant normalization, and the zero-based predictions of Heuristic 5.1.

```
import math, numpy as np
from sympy import primerange

FACT = [1]
for k in range(1, 501): FACT.append(FACT[-1] * k)

def ap_gauss(p):
    # a(p) for 32a1, exactly
    if p == 2 or p % 4 == 3: return 0
    for a in range(1, math.isqrt(p) + 1, 2):
        b2 = p - a * a; b = math.isqrt(b2)
        if b * b == b2:
            for s in (a, -a):
                for t in (b, -b):
                    if (s + t) % 4 == 1: return 2 * s

def D_sequence(h, N, c=1):
    """h -> j by (D1); prepend c; return D_m = m! h^(c)_m (
        exact ints)."""
    j = [0] * (N + 1)
    for m in range(1, N + 1):
        j[m] = m * h[m] - sum(j[r] * h[m - r] for r in range(1, m))
    jc = [0, c] + j[1:N]
    D = [0] * (N + 1); D[0] = 1
    for m in range(1, N + 1):
        ff, s = 1, 0
        # ff = (m-1)!/(m-r)!
        for r in range(1, m + 1):
            s += jc[r] * D[m - r] * ff
            ff *= (m - r)
        D[m] = s
    return D

def _mant_exp(num, den):
    # num/den = mant * 2**e,
    overflow-safe
    if num == 0: return 0.0, None
    sign = 1.0 if (num >= 0) == (den > 0) else -1.0
    num, den = abs(num), abs(den)
    e = num.bit_length() - den.bit_length()
    q = (num << 60) // (den << e) if e >= 0 else (num << (60 - e)) // den
    return sign * q * 2.0 ** -60, e

def mu_from_D(D, n, trust=(100, 22.0), fallback=(140, 38.0)):
    for M, R in (trust, fallback):
        M = min(M, n)
```



```

    me = [_mant_exp(D[n - m] * (-1) ** m, FACT[n - m] *
        FACT[m])
        for m in range(M, -1, -1)]
    E = max(e for _, e in me if e is not None)
    co = [0.0 if e is None else math.ldexp(mant, max(e - E,
        -1000))
        for mant, e in me]
    while len(co) > 1 and co[0] == 0.0: co = co[1:]
    keep = [abs(z) for z in np.roots(co) if abs(z) <= R]
    if keep: return min(keep)
    return float('nan') # runaway past the
        fallback radius

def smallest_zero(h): # of the truncation of H(
    t)
    co = np.array(h[::-1], dtype=float); co /= np.max(np.abs(co))
    z = min(np.roots(co), key=abs)
    return abs(z), abs(math.atan2(z.imag, z.real))

def predictions(h): # Heuristic 5.2
    r, th = smallest_zero(h)
    f = th / math.pi; fa = min(f, 1 - f)
    period = 1 / fa if fa > 1e-6 else float('inf')
    peak = math.pi / (2 * r * math.sin(th)) if 0.02 < th < math
        .pi - 0.02 \
        else float('inf')
    dips = 0.02 < th < math.pi - 0.02
    return dict(r=r, theta=th, dips=dips, period=period, peak=
        peak)

def controls(ps, h_real, rng): # synthetic sequences of
    Sec. 5.2
    N = len(ps)
    zero_here = [p == 2 or p % 4 == 3 for p in ps]
    raw = [2 * math.sqrt(p) * math.cos(t)
        for p, t in zip(ps, rng.uniform(0, math.pi, N))]
    to2 = lambda v: 4 * round((v - 2) / 4) + 2 #
        nearest 2 mod 4
    to0 = lambda v: (lambda m: m if m else 4)(4 * round(v / 4))
    to1 = lambda v: 2 * round((v - 1) / 2) + 1 #
        nearest odd
    out = {}
    out['A'] = [1] + [0 if z else int(round(v)) for z, v in
        zip(zero_here, raw)]
    out['A2'] = [1] + [0 if z else to2(v) for z, v in zip(
        zero_here, raw)]
    out['A0'] = [1] + [0 if z else to0(v) for z, v in zip(
        zero_here, raw)]
    out['A1'] = [1] + [0 if z else to1(v) for z, v in zip(
        zero_here, raw)]
    nz = [h_real[i + 1] for i in range(N) if not zero_here[i]]

```

```

    pos = set(int(i) for i in rng.permutation(N)[:sum(zero_here
    )])
    it = iter(nz)
    out['B'] = [1] + [0 if i in pos else next(it) for i in
    range(N)]
    out['C'] = [1] + [0 if rng.random() < 0.5 else int(round(v)
    ) for v in raw]
    return out

N = 300
ps = list(primerange(2, 10 ** 6))[:N]
h = [1] + [ap_gauss(p) for p in ps] # Figure 20B
    input
D = D_sequence(h, N, c=1)
mu = np.array([mu_from_D(D, n) for n in range(1, N + 1)])
print(predictions(h)) # r=0.7053, theta=2.4748, period 4.71,
    peak 3.60
# replication of Sec. 5.2: for seed in range(20), pass
# numpy.random.default_rng(seed) to controls(), run the
    pipeline on
# each of the six sequences, and record min(mu[100:]) and the
# dominant DFT period of mu[100:] - mean.

```

For Section 6, the coefficients $a(n)$ were built either from eta-quotient product expansions or multiplicatively from point counts:

```

def eta_expand(pairs, N):
    """q-expansion of prod_(m,e) eta(m tau)^e, exact integers.
    """
    shift = sum(m * e for m, e in pairs) // 24
    poly = [0] * (N + 1); poly[0] = 1
    for m, e in pairs:
        n = 1
        while m * n <= N:
            for _ in range(e): # multiply by (1
                - q^(mn))^e
                new = poly[:]
                for i in range(N + 1 - m * n): new[i + m * n]
                    -= poly[i]
                poly = new
            n += 1
    a = [0] * (N + 1)
    for i in range(N + 1 - shift): a[i + shift] = poly[i]
    return a
# 27a1: eta_expand([(3,2),(9,2)], N); 32a1: [(4,2),(8,2)]; 36a1
    : [(6,4)].
# For the other curves: a(p) by point counts on the stated
    models,
# a(p^k) by the Hecke recursion a(p^k)=a(p)a(p^(k-1))-p a(p^(k
    -2))
# (with a(p^k)=a(p)^k at bad p), extended multiplicatively;
    then

```

```
# smallest_zero(h) and the mu pipeline as above, and the tau_0
# of
# Lemma 6.1 is  theta/(2 pi) + i(-log r)/(2 pi).
```

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